

Simulation and Analysis of Secure Communication Using BB84 Quantum Key Distribution

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Abstract — This paper presents a proof-of-concept implementation of secure communication using Quantum Key Distribution (QKD) based on the BB84 protocol. The proposed system integrates quantum key generation with classical cryptographic techniques, where shared keys are derived using a Key Derivation Function (KDF) and applied for symmetric encryption using AES-GCM. A simulation using Qiskit is conducted to evaluate the protocol under intercept-resend eavesdropping attacks. The results demonstrate that the presence of an eavesdropper introduces a measurable Quantum Bit Error Rate (QBER) close to the theoretical expectations. Additionally, even when eavesdropping is not explicitly detected, it frequently causes key mismatches, leading to decryption failure. These findings highlight the probabilistic nature of eavesdropper detection in BB84 and demonstrate the effectiveness of combining quantum key distribution with classical encryption for secure communication.

Keywords—QKD, BB84, KDF, AES, Cryptography.

I. INTRODUCTION

Secure communication has become an essential technology in modern networks, especially if it handles critical information such as personal information, financial, and government data. To ensure security, researchers around the world have developed cryptographic systems like RSA and ECC using classical methods to solve mathematical problems that require huge computational power to break. However, the development of quantum technology will make these encryptions more vulnerable to attack. This problem has raised interest in exploration of quantum-safe communication methods.

Quantum Key Distributions (QKD) offers enhanced protection from attackers by leveraging the principles of quantum mechanics rather than computational complexity. One of the earliest and most popular QKD schemes is BB84 which was developed by Charles Bennett and Gilles Brassard in 1984. This protocol ensures security of communication between two parties where any attempt to eavesdrop the transmitted qubit will disturb the information, therefore revealing the existence of an eavesdropper.

Simulation-based analysis would be a valuable tool for understanding its practical behavior. A clear proof-of-concept implementation would help students and researchers visualize the implementation of its protocol and its robustness of security. That's why, in this work, we provide a software-based simulation and security analysis of the BB84 protocol. This paper analyzes: (1) implementation of a complete BB84 key-generation process including qubit preparation, measurement, basis revealing-reconciliation, and key creation for the two parties "Alice" and "Bob", (2) Introducing an eavesdropper "Eve" who attempt to intercept and measure the qubit resulting in changing of its information, and (3)

demonstrating how Alice and Bob can detect Eve presence and using its final key to encrypt message using Key Derivation Function (KDF) and AES encryption.

II. THEORETICAL BACKGROUND

A. Quantum Information Principles

Quantum information science relies on fundamental properties of quantum mechanics that distinguish it from classical information theory. In the context of Quantum Key Distribution (QKD), the following principles are paramount.

1. Superposition

Unlike a classical bit that must be in a state of either 0 or 1, a quantum bit (qubit) can exist in a linear combination of both states simultaneously. This phenomenon is known as superposition. For a qubit, this is mathematically expressed as:

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here, α and β are complex probability amplitudes. The fundamental condition is that the total probability must equal one, denoted by the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

This principle allows quantum protocols, such as BB84, to encode information in different bases (e.g., Rectilinear and Diagonal bases), creating the uncertainty required for secure transmission.

2. Born's Rule

In quantum mechanics, the act of measurement is not passive; it is governed by **Born's Rule**, which postulates that the probability of obtaining a specific measurement outcome is equal to the squared magnitude of the projection of the quantum state onto the corresponding basis vector. For a qubit in the superposition state $|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$, measured in the standard basis $\{|0\rangle, |1\rangle\}$, Born's rule dictates that the probability of observing outcome $|0\rangle$ is $P(0) = |\langle 0|\Psi\rangle|^2 = |\alpha|^2$ the probability of observing $|1\rangle$ is $P(1) = |\langle 1|\Psi\rangle|^2 = |\beta|^2$. These probabilities must satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$.

Upon measurement, the wave function discontinuously "collapses" to the observed basis state. Crucially, if a qubit prepared on one basis (e.g., rectilinear) is measured on a conjugate basis (e.g., diagonal), the projection amplitudes result in uniform probabilities (randomness). This disturbance allows the receiver (Bob) to detect the statistical anomalies introduced by an eavesdropper (Eve).

3. No Cloning Theorem

A critical pillar of quantum cryptography is the No-Cloning Theorem, formally proved by Wootters and Zurek in

1982. It states that it is impossible to create an independent and identical copy of an arbitrary unknown quantum state. This theorem guarantees the security of the BB84 protocol. Since an eavesdropper cannot clone the transmitted photons to measure them later, any attempt to intercept the key will inevitably introduce detectable errors. [1]

B. Property of Polarized Photons

Polarizing lights can be produced by sending an ordinary light beam through a polarizing filter. The light axis of polarization is determined by the orientation of its polarizing filter. Although polarization is a continuous variable, the uncertainty principle forbids measurements on any single photon from revealing more than one bit about its polarization. For example, if a light beam with polarization axis α is sent into a filter oriented at angle β , the individual photons behave dichotomously and probabilistically, being transmitted with probability of $\cos^2(\alpha - \beta)$. The photons only behave deterministic when they are parallel (fully certain transmission) and perpendicular (fully certain blocking) [2].

If someone might want to learn α information, they would need to measure it through another filter. A filter that is not perpendicular would result in probabilistic output and losing all its original information. Another way one might try to learn its information is through cloning an identical photon, but no-cloning theorem would prevent this [2].

The Hilbert space for a single polarized photon is 2-dimensional; therefore, the state of the photon can be described using linear combination using Dirac notation. For example, $|\rightarrow\rangle$ and $|\uparrow\rangle$ representing horizontal and vertical polarization (0 and 90°). This basis will be called rectilinear (R). Another basis for the same Hilbert space can use its diagonal (D) basis vectors, where it will represent 45° and 135° as $|\nearrow\rangle$ and $|\nwarrow\rangle$.

C. Quantum Key Distribution

Quantum Key Distribution (QKD) is a protocol that has been provably secure by which private key bits can be created between two parties over a public channel. The key bits will be used to implement a classical key cryptosystem, creating situations where both parties can communicate securely. The only requirement for the QKD protocol is that qubits can be communicated over the public channel with an error rate lower than a certain threshold. The security of the resulting key is guaranteed by the properties of quantum information and thus is conditioned only on fundamental laws of physics being correct.

The basic idea of QKD follows the fundamentals of quantum physics observation law where eavesdroppers can't gain any information without disturbing its state. Eavesdroppers also cannot clone the qubit due to the no cloning theorem.

The protocol exploits the transmission of non-orthogonal quantum states to detect channel interference. By monitoring disturbances in a randomly selected subset of 'check' qubits, Alice and Bob can statistically bound the level of noise or eavesdropping present in the transmission. This estimated error rate is assumed to represent the integrity of the data qubits as well. Subsequently, the parties employ information reconciliation and privacy amplification to extract a secure shared key. The maximum acceptable error threshold is ultimately defined by the capacity of these post-processing

algorithms to effectively correct errors and eliminate leaked information.

D. BB84 Protocol and Its Implementation

One user (Let's call her 'Alice') chooses a random bit string and a random sequence of polarization bases (rectilinear or diagonal). She will then send the other user (let's call him 'Bob') a train of photons, each representing one bit of the string in the basis chosen. She encodes bit 0 randomly for $|\rightarrow\rangle$ and $|\nearrow\rangle$, while encoding bit 1 with $|\uparrow\rangle$ and $|\nwarrow\rangle$. Bob will measure the photons he received whether they're rectilinear and diagonal thus measuring whether it's 0 or 1.

Bob will reveal their bases and photons. Depending on Bob's measurement choice, he chose the correct (diagonal) basis, and the state remains preserved; if he chose the wrong (rectilinear) basis, the outcome becomes random; if it had 90 degrees angle difference, it would be fully blocked.

Once the transmission is complete, Alice and Bob communicate via a public classical channel to perform the 'sifting' process. In this step, Bob announces which basis (rectilinear or diagonal) he used to measure each specific photon, and Alice replies by confirming whether that basis matched the one she used for encoding. Crucially, they do not reveal whether they sent/measured a 0 or a 1 (the bit values) at this stage. They only discuss the alignment of their measurement filters. Alice and Bob then sacrifice a small random subset of this Sifted Key to detect errors caused by eavesdropping. They publicly compare the bit values of this subset.

Since the eavesdropper (Eve) cannot clone the unknown quantum states, she is forced to measure them to gain information. Because she guesses the measurement basis randomly, she will choose the wrong basis 50% of the time, which causes the photon's state to collapse and change. When she re-sends this altered photon to Bob, he may measure an incorrect bit of value even if he and Alice used the matching basis. This process introduces a statistical error rate of approximately 25%, which alerts Alice and Bob to the breach.

E. Translating polarized photon into dirac notation

Horizontal polarization $|\rightarrow\rangle$ and vertical polarization $|\uparrow\rangle$ are physically distinguishable. If horizontal light being passed through a vertical polarizing, no light will get through. This perfectly matches the requirement for the computational basis states $|0\rangle$ and $|1\rangle$, where the inner product $\langle 0|1\rangle = 0$.

Therefore, we can arbitrarily but consistently map:

$$|\rightarrow\rangle \equiv |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |\uparrow\rangle \equiv |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

If you take a horizontal photon and rotate its polarization by 45°, classical physics says it is now diagonal. Quantum mechanics say it is in a **superposition** of horizontal and vertical. Mathematically, a 45° diagonal photon $|\nearrow\rangle$ is an equal mix of horizontal and vertical:

$$|\nearrow\rangle = \frac{|\uparrow\rangle + |\rightarrow\rangle}{\sqrt{2}}$$

This is mathematically identical to the definition of the $|+\rangle$ state in quantum computing, created by applying a Hadamard gate to $|0\rangle$:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Similarly, the anti-diagonal $|\searrow\rangle$ represents the **subtractive** superposition, corresponding to $|-\rangle$:

$$|\searrow\rangle \equiv |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Polarization	Dirac Notation
$ \rightarrow\rangle$	$ 0\rangle$
$ \uparrow\rangle$	$ 1\rangle$
$ \nearrow\rangle$	$ +\rangle$
$ \searrow\rangle$	$ -\rangle$

Table. 1: Photon polarization in Dirac notation

F. Qubit Measurement in Wrong Base

Measuring $|+\rangle$ and $|-\rangle$ with $|0\rangle$ and $|1\rangle$ as base will get probability as follows:

$$\begin{aligned} |0|+\rangle|^2 &= \frac{1}{2} & |1|+\rangle|^2 &= \frac{1}{2} \\ |0|-\rangle|^2 &= \frac{1}{2} & |1|-\rangle|^2 &= \frac{1}{2} \end{aligned}$$

If we were given a qubit that is $|+\rangle$ or $|-\rangle$ state, and we want to know whether the qubit is $|0\rangle$ or $|1\rangle$, we won't get any useful information. We can also express $|0\rangle$ and $|1\rangle$ in $|+\rangle$ and $|-\rangle$,

$$|0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}} \quad |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}$$

Measuring $|0\rangle$ and $|1\rangle$ with $|+\rangle$ and $|-\rangle$ as base (using diagonal polarization filter for vertically polarized photon) will get probabilities:

$$\begin{aligned} |+\rangle|0\rangle|^2 &= \frac{1}{2} & |-\rangle|0\rangle|^2 &= \frac{1}{2} \\ |+\rangle|1\rangle|^2 &= \frac{1}{2} & |-\rangle|1\rangle|^2 &= \frac{1}{2} \end{aligned}$$

If we choose wrong base to measure qubit, there is 50% chance of getting right answer and 50% chance we will get wrong answer, in other words, measuring qubits with wrong base will not give us any useful information.

III. DESIGN OF THE SECURE COMMUNICATION SCHEME

The scheme can be divided into three parts, secret key exchange, error detection, and message encryption.

In secret key exchange, Alice will send the qubits to Bob via quantum channel, then Bob will measure it into classical bits. In error detection, both Alice and Bob will publish some of their secret classical bits into classical channel, thus reducing the length of the shared secret key in exchange to determine if there is an eavesdropper. If there is no eavesdrop being detected, they will use the shared secret key bits that are still not being published as a secret key, then they will derive the encryption key using KDF to produce 256-bit length key for AES.

A. Key Exchange

The key exchange procedure starts by generating random bits K with length k , with k is the number of qubits that we use, we also need to generate random bits with length k as our bases to encode the shared key, here we will use 0 as rectilinear base and 1 as diagonal base. In rectilinear base, 0 will be encoded as $|0\rangle$ and 1 will be encoded as $|1\rangle$, in diagonal base 0 will be encoded as $|+\rangle$ and 1 will be encoded as $|-\rangle$.

In Qiskit, every qubit is initialized as $|0\rangle$, to make the qubit as $|1\rangle$, we just need to apply Pauli-X or NOT gate, to convert qubit from $|0\rangle$ to $|+\rangle$, we apply Hadamard gate, then to produce $|-\rangle$, we apply $|0\rangle$ to Pauli-X gate, then Hadamard gate.

After encoding K into qubits, we then transmit the qubits to the receiver. The receiver then will generate random k bits as measurement base, 0 for rectilinear base, 1 for diagonal base. Then the receiver will perform measurements. If the measurement base is 0, then the receiver will perform the measurement right away, if the measurement base is 1, then they will apply Hadamard gate to the qubit, then measure it.

Remember that if the transmitted qubit is $|+\rangle$, it means that the $|0\rangle$ qubit is going through a Hadamard gate, which means that applying Hadamard gate for the second time will mean we make a cascade of two Hadamard gate, a cascade of two Hadamard gate is an identity gate, so it means that we get back to the original $|0\rangle$ qubit.

If the transmitted qubit is $|-\rangle$, it means that the $|0\rangle$ qubit is going through a Pauli-X gate (thus, become $|1\rangle$), then Hadamard gate, applying Hadamard gate for the second time will mean we make A cascade of two Hadamard gate, so we left with Pauli-X gate. Since every qubit is initialized as $|0\rangle$, a Pauli-X applied to the initial qubit will produce qubit $|1\rangle$.

If the transmitted qubit is either $|0\rangle$ or $|1\rangle$, applying Hadamard will result in $|+\rangle$ or $|-\rangle$, measuring these qubits will give us 0 or 1 with each having 50% probability.

Then, both sender and receiver will publish their bases, if they are matching bases, then we will discard the measurement result. To illustrate an example, consider following sequences, taken from [4]:

$$\mathbf{k} = 11000100, \quad \mathbf{a} = 11010011, \quad \mathbf{b} = 01011001$$

$\mathbf{k}$ is the randomly generated bits, $\mathbf{a}$ is the bases chosen by Alice, $\mathbf{b}$ is the bases chosen by Bob.

Initial bit sequence $\mathbf{k}$	1 1 0 0 0 1 0 0
Alice encodes:	$\nwarrow \nwarrow \rightarrow \nearrow \rightarrow \uparrow \nearrow \nearrow$
Bob measures:	$\uparrow \nwarrow \rightarrow \nearrow \nwarrow \uparrow \rightarrow \nearrow$
Bases comparison:	$\times \checkmark \checkmark \checkmark \times \checkmark \times \checkmark$
Shared key $\tilde{\mathbf{k}}$	1 0 0 1 0

Table. 2: Example of BB84 without eavesdropping

When Bob performs his measurement, the result is colored blue to indicate that it is probabilistic. As it can be noticed, out of all three wrong guesses he took, the first and last states he

observes are indeed the correct encodings of the bits in the rectangular basis, but he only owes this to chance. In the end, they arrive at a 5-bit shared key.

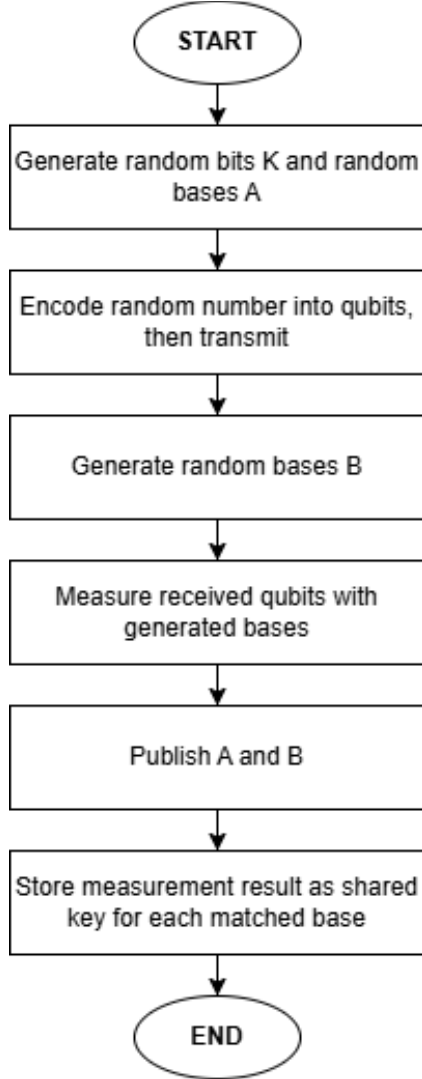


Fig. 1: Flow chart of BB84 without eavesdropping

B. Error Detection

Because of no-cloning theorem, to get the transmitted qubits, the eavesdropper needs to measure the qubit, then re-encode it into qubits based on their measurement and their chosen bases. Since the eavesdropper did not know the bases chosen by sender, generating bases to measure these transmitted qubits is necessary. Inevitably, this action might alter the state of the qubit. Let's identify several possible eavesdropping scenarios, let Alice send the message to Bob, and Eve is an eavesdropper.

- *Eve Chooses the wrong basis:* $e_i = \bar{a}_i \rightarrow$ qubit is altered.
- *Bob chooses correctly:* $b_i = a_i \rightarrow$ Eve introduces error with 50% probability.
- *Bob chooses incorrectly:* $b_i = \bar{a}_i \rightarrow$ random outcome, Eve is undetected.
- *Eve chooses the correct basis:* $e_i = a_i \rightarrow$ qubit is unaltered.

- *Bob chooses correctly:* $b_i = a_i \rightarrow$ Eve is undetected and has one bit of the key.
- *Bob chooses incorrectly:* $b_i = \bar{a}_i \rightarrow$ random outcome, Eve is undetected.

These possibilities reveal that for each transmitted qubit, there is 75% probability that Eve's action goes undetected. The remaining 25% probability is due to Bob's correct choice when Eve chooses incorrectly: he obtains the wrong state.

To check if there is an eavesdropper, both Alice and Bob will publish some of their keys to catch Eve. After publishing their bases (A and B), Alice (sender) will publish a random number as a seed to a Random Number Generator (RNG), the random number will be used to choose which bit in the shared key that will be published by Alice and Bob. We use RNG so that Eve did not know in advance which bits we are going to use as detection bits. If Eve were to know in advance which bits are selected as detection bits, she could avoid measuring them, thereby evading the detection mechanism implemented by Alice and Bob. Detection bits are some bits from agreed shared key that is being published to the public to detect any eavesdropping attempt (we sacrifice some of our bits, reducing the length of our shared key). If at least one bit were mismatched, we know there is an attempt to intercept the communication. If one bit that Alice and Bob publish is mismatched, we know that someone is altering the bits, thus the protocol is aborted.

Let's take a look at one example taken from [4]. Sequences k_A and k_B belonging to Alice and Bob respectively are "distilled" from the initial k . Eve's bases are represented by e .

$k_A = 11000100$, $a = 11010011$, $b = 01011001$, $e = 10001001$

Initial bit sequence k	1	1	0	0	0	1	0	0
Alice encodes:	$\nwarrow$	$\nwarrow$	$\rightarrow$	$\nearrow$	$\rightarrow$	$\uparrow$	$\nearrow$	$\nearrow$
Eve measures:	$\nwarrow$	$\uparrow$	$\rightarrow$	$\rightarrow$	$\nearrow$	$\uparrow$	$\rightarrow$	$\nearrow$
Bob measures:	$\uparrow$	$\nearrow$	$\rightarrow$	$\nearrow$	$\nearrow$	$\uparrow$	$\rightarrow$	$\nearrow$
Bases comparison:	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$
Alice's bit sequence k_A	1	0	0			1	0	
Eve's information:	1	0				1	0	
Eve introduces error?	N	Y	N	N	N	N	N	N
Bob's bit sequence k_B	0	0	0			1	0	

Table.3: Example of BB84 with eavesdropping

From Eve's choices she guessed the bases correctly four out of eight times, thus she gains information about four bits from the initial bit sequence. She introduces an error in the second bit, making Bob decode 0 instead of 1.

In a real scenario, Alice and Bob would need to compare many bits, from k_A and k_B , which are about $n/2$ bits long on average, for n is the number of bases chosen to sender or receiver. If both compare one bit from their respective bases, the probability of them being the same (p_e) is 0.75. For a selection of $\tilde{n}$ bits, the probability of having all of them matches represents Eve's chance of evading detection, which

decreases exponentially with $\tilde{n}$. Therefore, the probability of detection p_d that Alice and Bob wish to have.

$$p_d = 1 - p_e = 1 - \left(\frac{3}{4}\right)^{\tilde{n}}, \tilde{n} < \frac{n}{2}.$$

For example, $\tilde{n} = 20$ making the probability of finding the eavesdropper $p_d \approx 0.997$. The probability of finding the eavesdropper is depicted below.

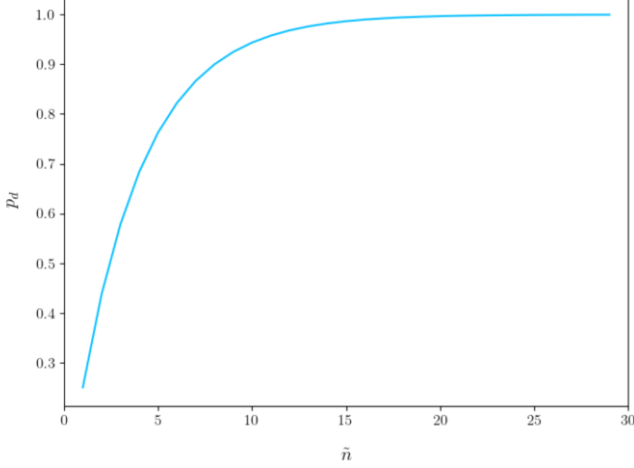


Fig. 2: Probability of detecting Eve increases with $\tilde{n}$

C. Message Encryption

After both parties get shared key and make sure that nobody is eavesdropping, they will feed the shared key into KDF, here we choose HKDF to generate fixed random key for AES. Due to Grover's algorithm, that allows to perform an exhaustive search in the square root of the classical time, primitives providing 128-bits of security against quantum adversaries need to have a key length of at least 256 bits. That is the main reason why the actual recommendation for encryption with post-quantum security is the version of AES with a 256-bit key [3].

For the encryption, we choose AES-GCM scheme, because it provides both confidentiality and authentication, this scheme also lets us have additional authenticated data that can be sent in clear text. This means, not only impossible to see the original message, it's also impossible alter the message.

Because we need matching bases between both parties to append 1 bit to the shared key, it means that there is 50% probability that both parties are using the same base for one qubit, which means it makes sense that we consider we will lose half of our transmitted qubits. We also need to publish some of our shared keys as detection bits to detect errors caused by eavesdropping, this reduces our shared key even more; by looking at Fig. 2, it is considered enough to have at least 30 bits. In practical implementations, significantly larger numbers of qubits are required to ensure reliable detection and sufficient key length.

IV. EXPERIMENTAL RESULTS

To evaluate the proposed BB84-based secure communication scheme, we conducted simulation-based experiments using Qiskit. Due to the computational limitations of classical hardware, particularly the exponential complexity of simulating quantum states, the number of qubits used in the simulation is restricted to fewer than 26 qubits.

Additionally, for demonstration purposes, a portion of the shared key is revealed as detection bits to emulate the error detection phase, while the remaining bits are preserved as the secret key.

Figures 3–6 illustrate several representative scenarios observed during the simulation, including both secure communication and various eavesdropping conditions.

```
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication> python bb84_qkd.py 20 --eavesdrop 0
Matching basis ratio: 0.5
Alice key == Bob key: True
Ciphertext: 861e437892fada1fb38953c31a024ca4e4a8345ba4db912ae9cec84fb7cc
Decrypted message: Secret Message
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication>
```

Fig. 3: A normal communication scenario without the presence of an eavesdropper.

In this case, Alice and Bob successfully generate identical shared keys and are able to perform encryption and decryption without errors.

```
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication> python bb84_qkd.py 20 --eavesdrop 1
Matching basis ratio: 0.55
Detection bit altered: Eavesdropper detected! Aborting key generation
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication>
```

Fig. 4: Eavesdropping attempt is detected.

In this case, at least one detection bit differs between Alice and Bob, indicating disturbance in the quantum channel. As a result, the protocol correctly aborts the key generation process.

```
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication> python bb84_qkd.py 20 --eavesdrop 1
Matching basis ratio: 0.5
Alice key == Bob key: True
Ciphertext: 1f07b350d44922bc8d21a2667fe031d9776f8c4da4b82b8c582f243451c5
Decrypted message: Secret Message
WARNING: Eve exists but no bits were altered!
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication>
```

Fig. 5: Eavesdropping attempt does not affect the selected detection bits, but alters the remaining key bits.

Consequently, Alice and Bob derive different keys, leading to decryption failure. However, since the detection subset does not capture the disturbance, the presence of the eavesdropper is not explicitly identified.

```
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication> python bb84_qkd.py 20 --eavesdrop 1
Matching basis ratio: 0.45
Alice key == Bob key: True
Ciphertext: 1f07b350d44922bc8d21a2667fe031d9776f8c4da4b82b8c582f243451c5
Decrypted message: Secret Message
WARNING: Eve exists but no bits were altered!
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication>
```

Fig. 6: Eavesdropping attempt does not introduce observable discrepancies in either the detection bits or the final key (Rare Scenario).

This occurs due to the probabilistic nature of quantum measurements, where Eve may coincidentally choose the correct measurement bases or the introduced disturbances are not reflected within the sampled subset. It is important to emphasize that this phenomenon does not violate the security guarantees of the BB84 protocol. Rather, it reflects the inherent probabilistic detection mechanism, where the likelihood of undetected eavesdropping decreases exponentially with the number of detection bits. This behavior is consistent with the probability term $(3/4)^{\tilde{n}}$, which represents the chance that all sampled bits remain consistent despite the presence of an eavesdropper.

To further evaluate the performance of the system, we conducted multiple independent simulation runs. Specifically, we performed 100 runs using 20 qubits per run under an intercept-resend attack model.

```
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication> python simulation.py 20 --runs 100 --eavesdrop 1
=== RESULT ===
Runs: 100
Bits: 20
Eve enabled: True
Average QBER: 0.29
Detection Rate: 0.48
Mismatch Rate: 0.91
PS C:\Users\andhi\Documents\GitHub\BB84_Secure_Quantum_Communication>
```

Fig. 7: Simulation of 100 runs using 20 qubits per run under an intercept-resend attack model.

The simulation results show that the presence of an eavesdropper introduces a significant error rate, with an average QBER of approximately 0.29. This value is close to the theoretical expectation of 25% for intercept-resend attacks, with the deviation attributed to finite sample size and statistical fluctuations in the simulation.

The detection rate of 48% indicates that the eavesdropper is identified in nearly half of the simulation runs. This moderate detection rate is primarily due to the limited number of qubits and the relatively small subset of bits used for QBER estimation, which reduces the likelihood of capturing all disturbances introduced by the eavesdropper.

Furthermore, the key mismatch rate is high (91%), indicating that even when the eavesdropper is not explicitly detected, their interference frequently disrupts the key agreement process. As a result, Alice and Bob often derive different keys, leading to decryption failure. This behavior demonstrates an additional layer of implicit security, where undetected eavesdropping still leads to unsuccessful communication due to key inconsistency.

To further illustrate the statistical behavior of eavesdropper detection, we analyze the detection rate as a function of the number of simulation runs, as shown in Fig. 8.

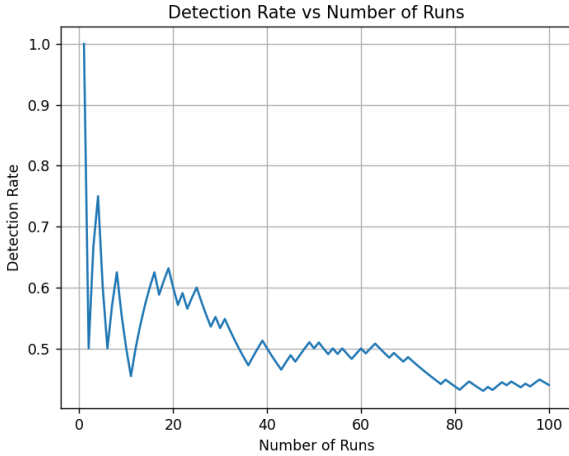


Fig. 8: Detection rate as a function of the number of simulation runs.

As shown in Fig. 8, the detection rate initially exhibits significant fluctuations due to the randomness of individual simulation runs and the small sample size. However, as the number of runs increases, the detection rate gradually stabilizes and converges to a value around 0.40–0.50.

This convergence behaviour demonstrates the statistical nature of eavesdropper detection in the BB84 protocol. While

detection is not guaranteed in each individual run, repeated executions reveal a consistent detection probability.

The stabilization of the detection rate also supports the reliability of the experimental results, indicating that the observed detection rate of approximately 48% reflects the underlying probabilistic behavior of the protocol rather than random variation. Furthermore, increasing the number of runs improves the accuracy and stability of detection probability estimation.

In practical QKD systems, significantly larger key sizes and more extensive sampling are employed, along with additional processes such as error correction and privacy amplification, to ensure that any partial information obtained by an eavesdropper is effectively eliminated.

The source code for the simulation is available [here](#)

V. CONCLUSIONS

In this work, we presented a proof-of-concept implementation of secure communication using the BB84 Quantum Key Distribution (QKD) protocol. By leveraging fundamental quantum properties such as the no-cloning theorem and measurement disturbance, the system enables the detection of eavesdropping through observable error rates. Simulation results demonstrate that the presence of an eavesdropper introduces a measurable QBER close to theoretical expectations. Additionally, even when not explicitly detected, eavesdropping frequently leads to key mismatches, resulting in decryption failure. This behavior provides an additional layer of implicit security, where successful undetected attacks remain unlikely in practice. These findings highlight the probabilistic nature of eavesdropper detection in BB84, where reliability improves with larger key sizes and increased sampling. In practical systems, additional processes such as error correction and privacy amplification are required to further strengthen security guarantees. Future work includes integrating noise models and implementing full BB84 post-processing steps such as error correction and privacy amplification.

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